Recalibrating Visual GEO: An Anchor-Decoupled Evaluation Protocol and a Cross-Encoder Transfer Probe on ESCI

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Abstract*—Visual generative engine optimization (VGEO) edits a product image so it ranks higher under a CLIP-style retriever, and headline results report large cosine-similarity gains at high SSIM, arguing that visual GEO is a serious retrieval-marketplace threat. We show this is largely a measurement artifact: when the evaluation query is itself derived from the optimization anchor, any such gain is structurally inflated rather than a measure of held-out retrieval impact. We make two contributions. First, an anchor-decoupled evaluation protocol — pinned by a single anchor⊥query invariant ($v_{\mathrm{anchor}}(x_p)$ depends only on x_p), a pre-flight retriever-discrimination gate, a per-cohort decomposition over the four ESCI human-graded relevance classes, and a compact reference verifier (released as HONESTVGEO) — that re-measures the white-box threat rather than amplifying it. Second, AE-CoGEO, to our knowledge the first query-free, anchor-decoupled, cross-encoder-consensus transfer attack for visual GEO, which doubles as the probe that makes the transfer axis measurable. On the OpenAI ViT-L/14 white-box axis CoGEO leads the four-way 560-pair intersection on aggregate rank-lift ($+19.86$, paired Wilcoxon $p \leq 1.1 \times 10^{-4}$), and the lead replicates as the unique top-1 on five of six retrievers. Yet this lead is concentrated in the harder relevance cohorts rather than being a per-pair median advantage, trades off roughly monotonically with visual stealth, and — under the strongest ensemble-transfer attack we could build — stays far below the white-box ceiling of the same source-consensus perturbation. Read honestly, visual GEO is a cohort-specific, white-box-bounded, transfer-limited, and detectable (AUROC $\geq$ 0.99 at ε=$16/255$ on a balanced benchmark, lower at smaller budgets) operator, not a uniform invisible threat.

***Index Terms*—visual retrieval, CLIP, adversarial perturbation, generative engine optimization, Amazon ESCI, multimodal evaluation**

I. INTRODUCTION

Visual generative engine optimization (VGEO) is the seller-side analogue of search-engine optimization for retrieval systems that match natural-language queries to product images through a shared multimodal embedding [1], [2]. Imperceptible perturbations of a product image can substantially change its rank under a CLIP-style retriever [3]–[5], and headline results report large positive Δ in cosine similarity together with high SSIM [6], arguing that visual GEO is a real threat to retrieval marketplaces.

The conventional evaluation protocol for visual GEO is unsound in a reproducible way: the eval-time query is derived from the same anchor that drove the optimization. When $v_{\mathrm{anchor}}(x_p)$ is the CLIP encoding of a description of x_p and the eval-time similarity is measured against the same caption, any positive Δ is a tautology of gradient ascent: a measure of optimization vigor, not of retrieval impact under a real shopper query. The eval-time query must therefore be *decoupled* from the optimization anchor.

We deliver *HonestVGEO*, a reusable anchor-decoupled protocol that recalibrates visual GEO into an actionable, defender-relevant threat characterization rather than a deflated headline number. Figure 1 sketches the pipeline: a seller-side anchor depending only on x_p feeds four white-box attack families, while a held-out ESCI shopper query enters only at evaluation and never crosses the optimization gradient. The protocol pins three commitments: an *anchor⊥query invariant* making the anchor a function of the product image alone, checked by a compact ∼60-line reference verifier (released artifact); an *ESCI-grounded benchmark* [7] of 491 product images balanced across the four ESCI classes (Exact, Substitute, Complement, Irrelevant) with 560 paired triples, replicated across six CLIP backbones; and a *four-way comparison* of CoGEO, PGD-bare [8], AdvCLIP [4], and Co-Attack [5] at $\varepsilon = 16/255$, with per-cohort decomposition and a paired Wilcoxon matrix [9]. To reach the second, deployment-realistic transfer axis the white-box protocol cannot, we contribute *AE-CoGEO* (§III-D), the first query-free, anchor-decoupled cross-encoder transfer probe: it optimizes one perturbation against a source-encoder consensus to generalize to an unseen deployed encoder, giving defenders a concrete operator to measure against.

Three claims survive the protocol. **(C1)** Under the anchor-decoupled protocol, CoGEO is the strongest of the four compared CLIP-targeted attacks on the white-box axis: $+19.86$ mean rank-lift on the four-way intersection, with paired Wilcoxon $p \leq 1.1 \times 10^{-4}$ against every alternative (an aggregate-mean effect carried by the harder cohorts, not a uniform per-pair win over PGD-bare; §VI-E) — not a state-of-the-art claim. **(C2)** Aggregate dominance hides per-cohort

Code, the ESCI manifest, the no-model admissibility verifier, the reference rank-lift scorer, reference implementations of both attacks, reference result tables, and the full reproduction configuration (sampling, seeds, hyperparameters, hardware) are released anonymously as HONESTVGEO: https://anonymous.4open.science/r/honestvgeo-F663.

heterogeneity: CoGEO leads by large margins on Complement (+36.8) and Irrelevant (+25.4); on the Exact cohort the mean rank-lift gap reverses (+3.9 for CoGEO vs. +7.1 for PGD-bare), but matched-pair Wilcoxon and sign tests do not reject the null, so the two methods are statistically balanced rather than dominated. Co-Attack additionally turns negative on Complement, so single-number visual-GEO benchmarks conceal the per-cohort operating regime. **(C3)** Visual stealth and retrieval lift trade off across attack families: the SSIM means rank in broadly the reverse order from rank-lift (AdvCLIP 0.78 > PGD-bare/Co-Attack 0.76 > CoGEO 0.66), and CoGEO's SSIM ≥ 0.98 holds at 800×800 but not at the 224×224 regime that industrial CLIP retrieval consumes [1].

The boundary against prior work is twofold. On the evaluation side, any visual-GEO author can run the verifier of §III-A before reporting rank-lift, and the multi-method, multi-backbone replication [10]–[12] establishes the per-cohort pattern as a property of the attack family, not any single embedding. On the attack side, AE-CoGEO is, to our knowledge, the first query-free, anchor-decoupled, cross-encoder-consensus transfer attack for visual GEO (Table I). We assume a seller-side attacker with read access to one or more *publicly available* CLIP encoders and unlimited per-image gradient compute, lifting an uploaded perturbation's rank against held-out shopper queries on the same encoder (white-box axis) or an unseen, privately deployed one (transfer axis). A full defensive characterization is beyond our scope, but the transfer measurement (§VI-G) and a training-free detector already characterize visual GEO as a cohort-specific, white-box-bounded, transfer-limited operator.

II. Related Work

Generative engine optimization: Generative engine optimization was introduced for text-only generative search engines [2], where small content edits change which sources a system cites, sharing roots with SEO and adversarial information retrieval [13]. Visual GEO extends it to product retrieval ranked by a multimodal embedding rather than a generative LM [1], [14]. Our CoGEO attack combines a VLM-derived anchor, a Sobel mask, a differentiable JPEG layer [15], and an MI-FGSM loop [16]; it is one of four compared methods, with a VLM captioner [17], [18] supplying the decoupled anchor text.

CLIP-targeted adversarial attacks: Universal adversarial patches [4], [19] on a CLIP image encoder shift retrieval across diverse inputs, and a complementary family attacks both image and text branches with alternating updates [5]; both are query-agnostic, matching the seller-side regime where uploads outpace per-image budgets. The per-image branch descends from PGD [8] and FGSM [3] with momentum [16], and recent work maps the vulnerability surface of contrastive pretraining [20], [21] and robust-CLIP defenses [22]. What is missing is evaluation under held-out human relevance: prior work scores against queries derived from the optimization signal itself, never a multi-cohort benchmark with a published anchor⊥query invariant.

TABLE I
WHERE AE-COGEO SITS AMONG ADVERSARIAL ATTACKS ON THE FOUR AXES VISUAL GEO IMPOSES. **QF**: QUERY/CAPTION-FREE ATTACK; **HE**: SCORED UNDER HELD-OUT HUMAN RELEVANCE (ANCHOR⊥QUERY); **XE**: TRANSFERS TO AN UNSEEN ENCODER VIA SOURCE-SET CONSENSUS; **VG**: OPTIMIZES RETRIEVAL RANK, NOT CLASSIFICATION. AE-COGEO UNIQUELY SATISFIES ALL FOUR; **HE** HOLDS ONLY FOR OUR PROTOCOL.

Attack	QF	HE	XE	VG
PGD / FGSM [3], [8]	✓	×	×	×
MI/DI/SI-FGSM [16], [23], [24]	✓	×	×	×
Ensemble transfer [13]	✓	×	✓	×
AdvCLIP patch [4]	✓	×	×	✓
Co-Attack [5]	×	×	×	✓
CoGEO (ours)	✓	✓	×	✓
AE-CoGEO (ours)	✓	✓	✓	✓

Transfer-based attacks and cross-encoder consensus: A long line improves the black-box transferability of per-image attacks through input-diversity [23] and scale-invariant [24] transformations, momentum [16], and optimization against an ensemble of surrogate models [13], so a perturbation crafted on accessible models survives on an unseen one. These techniques were developed for image classification and still presume a target label or a query/caption at attack time [5]. AE-CoGEO (§III-D) is, to our knowledge, the first to bring ensemble-consensus transfer to visual GEO under the query-free, anchor-decoupled constraint, measuring rank-lift on held-out encoders that never entered the optimization set (§VI-G). Table I positions it among these families.

Held-out relevance evaluation: The Amazon ESCI dataset [7] provides (q, x_p, ℓ) triples with four-class human labels (Exact, Substitute, Complement, Irrelevant), and to our knowledge has not previously been used to evaluate adversarial image perturbations; we add a Food-101 [25] consistency check. Because ESCI queries come from real shoppers and labels are independent of any retriever's geometry, observed rank-lift measures something other than the gradient that produced it, placing the protocol in the held-out tradition of BEIR [26] and MTEB [27] that rejects self-loop similarity [22], [28]. Per-cohort decomposition is a first-class output, not a post-hoc breakdown.

Stealth and replication: SSIM [6] is the canonical perceptual-quality measure, but visual-GEO stealth claims are often reported at the optimization resolution (800×800) rather than the 224×224 input the retriever consumes — a discrepancy not previously characterized across attack families on one benchmark. We report the four-method stealth–lift Pareto at native input size and replicate across six CLIP backbones [1], [10]–[12], separating embedding-specific dominance from dominance over per-image gradient attacks across CLIP-style embeddings.

III. Anchor-Decoupled Evaluation Protocol

A. The anchor⊥query invariant

Let x_p denote a product image, q a shopper query in natural language, E_I and E_T the image and text encoders of a CLIP-style retriever [1], [10], [14], and $v_{\text{anchor}}(x_p) \in \mathbb{R}^d$ a target text embedding used to drive a perturbation δ such that $E_I(x_p+\delta)$ moves toward $v_{\text{anchor}}(x_p)$.

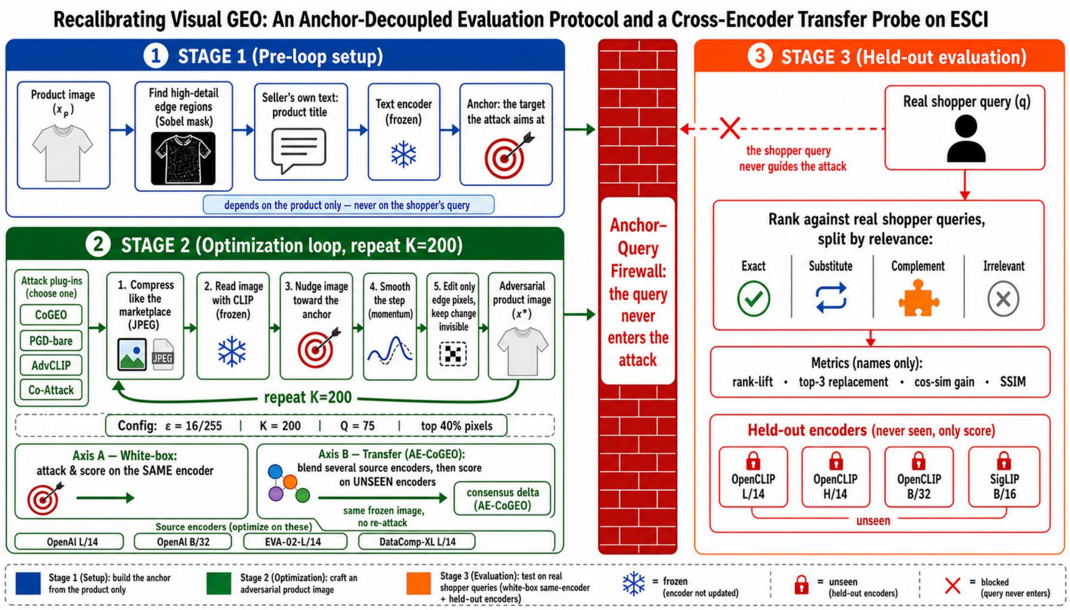

Fig. 1. Anchor-decoupled evaluation protocol with a cross-encoder transfer probe, instantiated for CoGEO. **Stage 1** precomputes the seller anchor $v_{\text{anchor}}{=}E_T(t_p^\star)$ from the product title and a Sobel-edge mask M_E (top-40% gradient pixels); v_{anchor} depends on the product image alone, never on a shopper query. **Stage 2** runs a five-step inner loop (differentiable JPEG $Q{=}75$, frozen image encoding, anchor-alignment cosine loss, MI-FGSM momentum, masked ℓ_∞ projection) for $K{=}200$ at $\varepsilon{=}16/255$, shared by PGD-bare, AdvCLIP and Co-Attack. Two orthogonal axes, not competing attacks: *Axis A* optimizes against a single white-box encoder; *Axis B* (AE-CoGEO) against a four-source consensus loss $L_{\text{ens}}{=}-\frac{1}{|S|}\sum_j \cos(E_I^{(j)}(\tilde{x}), E_T^{(j)}(t_p^\star))$. **Stage 3** scores the frozen image against unseen ESCI shopper queries across the four cohorts (E/S/C/I) and re-scores Axis B—no re-attack—on four held-out encoders. The shopper query never crosses the Anchor-Decoupled Query Firewall. Snowflakes mark frozen modules.

We require that, for every (q, x_p) pair used in evaluation,

$$v_{\text{anchor}}(x_p) \text{ is a function of } x_p \text{ alone, not of } q. \tag{1}$$

Concretely, v_{anchor} may be the CLIP text encoding of the product category, of a VLM description of x_p (varied across BLIP, BLIP-2, LLaVA-1.5, Qwen2-VL, and Qwen2.5-VL in the released artifact), or of the product title; in all cases it depends only on the seller-side description. The query q enters only at evaluation time, when we measure rank-lift, top-3 replacement, and held-out cosine similarity gains under q.

This invariant is the operational definition of an honest VGEO evaluation: when the eval query is itself derived from v_{anchor}, the similarity gain is structurally inflated relative to held-out retrieval impact. We enforce (1) at the code level across all four families with a compact reference verifier (~60 lines, released artifact) that statically rejects any query-like attack parameter and dynamically checks that the perturbation is byte-identical across swapped eval queries under a fixed seed.

B. Pre-flight monotonicity gate

Before any adversarial perturbation, the chosen retriever must demonstrate that human-graded relevance labels are discriminable in its embedding space. Writing $\overline{\cos}_\ell = \text{mean}_{(q,x_p)\in\ell}\big[\cos(E_I(x_p),\, E_T(q))\big]$ for the mean no-perturbation cosine over a cohort ℓ, we require that on a balanced sample of (q, x_p, ℓ) triples spanning the four ESCI relevance classes Exact (E), Substitute (S), Complement (C), Irrelevant (I), the gate

$$\overline{\cos}_{\ell=E} - \overline{\cos}_{\ell=I} \;\geq\; 0.05 \tag{2}$$

holds. A retriever that fails (2) cannot serve as an evaluation substrate: no adversarial pressure is measurable against an embedding that cannot already separate Exact-match from Irrelevant pairs. We read 0.05 as a qualitative admissibility floor, not a calibrated statistic: a conservative margin that the canonical CoGEO retriever (ViT-L/14, gap 0.0533) clears and a markedly weaker substrate (ViT-B/32, 0.0405) does not, with both satisfying $E > S > C > I$ monotonicity. The evaluation substrate (ViT-L/14) is fixed independently as the canonical retriever, so no downstream result depends on the exact floor value. On failure, the protocol mandates a backbone swap before optimization.

C. Four-way attack pass

Within the invariant, four white-box attack families are evaluated at matched perturbation budget $\varepsilon = 16/255$:

CoGEO: Per-image projected gradient descent [3], [8] on x_p with momentum (MI-FGSM [16]), Sobel-edge mask restricting perturbation to high-gradient regions, environment simulation through a differentiable JPEG layer [15] together with input-diversity [23] and scale-invariant [24] transformations, and product-title v_{anchor}; CoGEO is the attack we propose and the primary subject of the held-out evaluation. This recalibrated instantiation differs from prior CoGEO in three respects: a hard top-40% Sobel-gradient mask (vs. the original soft min-max-normalized mask floored at 0.1), differentiable JPEG at $Q{=}75$ (vs. $Q{=}85$), and translation-invariant (TIM) smoothing omitted, as it gave no significant lift under the anchor⊥query setting.

Formally, let $x_p^{(k)}$ denote the listing image at iteration k (with $x_p^{(0)}{=}x_p$). The Sobel edge mask

$$M_E \;=\; \mathbf{1}\big[\,|\nabla_{\text{Sobel}}\, x_p| \;\geq\; q_{0.6}(|\nabla_{\text{Sobel}}\, x_p|)\,\big] \qquad (3)$$

selects the top-40% gradient pixels (threshold $q_{0.6}$, the 60-th percentile of edge magnitude) and is computed once per image. At each iteration k, CoGEO applies a differentiable JPEG layer at quality $Q{=}75$ to make the attack compression-aware, embeds the result with the frozen CLIP image encoder, and forms the anchor-alignment loss

$$L^{(k)} \;=\; -\,\cos\!\big(\,E_I(\text{diffJPEG}_{Q=75}(x_p^{(k)})),\; v_{\text{anchor}}\,\big). \qquad (4)$$

The signed gradient is smoothed by a momentum buffer with $\mu = 1.0$ following the original MI-FGSM formulation [16]:

$$g^{(k+1)} \;=\; \mu\, g^{(k)} \;+\; \frac{\nabla_x\, L^{(k)}}{\|\nabla_x\, L^{(k)}\|_1}, \quad \mu = 1.0, \qquad (5)$$

and, since maximizing anchor cosine is gradient descent on $L^{(k)}$, the masked signed-momentum step (step size $\alpha = 4/255$) is projected into the ℓ_∞ ε-ball around the clean image:

$$x_p^{(k+1)} \;=\; \Pi_{\|\,\cdot\,-x_p\|_\infty \leq \varepsilon}\Big(\, x_p^{(k)} \;-\; \alpha \cdot \text{sign}\big(M_E \odot g^{(k+1)}\big)\,\Big). \qquad (6)$$

After $K{=}T{=}200$ iterations, $x_p^\star$ is the adversarial image evaluated against the held-out query in §III-E.

PGD-bare: The same product-title v_{anchor} as CoGEO but without the Sobel mask, without environment simulation, and without momentum, recovering the classical signed-step PGD baseline of [8]. PGD-bare isolates the contribution of the anchor from the engineering machinery layered on top.

AdvCLIP: A query-agnostic universal adversarial patch [19], [29] trained on the CLIP image encoder [4] over 30 epochs with budget $\varepsilon = 16/255$, applied to each x_p at evaluation. AdvCLIP requires no per-image gradient pass at deployment time and represents the universal-patch family of CLIP-targeted attacks.

Co-Attack: A cross-modal alternating attack [5] that interleaves an L_∞ image perturbation ($\varepsilon = 16/255$) and an L_2 text-anchor perturbation ($\varepsilon_v = 0.1$) over 200 iterations. The text branch operates on the seller-side anchor only, never on q, so the invariant is preserved.

All four pipelines take only x_p as optimization input; the held-out ESCI query never enters any optimization gradient, verified at the code level.

The CoGEO inner loop is fully specified by Eqs. (3)–(6); the full evaluation-pass pseudocode is in the released artifact.

D. AE-CoGEO: a cross-encoder-consensus transfer attack

The four-way comparison above is a *white-box* measurement: the encoder that drives the perturbation also scores it. This bounds one axis — how far an attacker can move the ranking of a retriever it fully controls — but leaves unmeasured the realistic deployment axis, where the platform retriever is an *unseen* encoder the attacker never optimized against. We introduce **AE-CoGEO** (Anchor-decoupled Ensemble CoGEO) to measure this second axis.

AE-CoGEO keeps both protocol commitments of CoGEO unchanged — the perturbation is *query-free* and *anchor-decoupled*, so the held-out shopper query never enters optimization ((1)) — and adds a single mechanism: instead of aligning to one encoder, it optimizes one perturbation against the *consensus* of a source set of encoders $\mathcal{S} = \{(E_I^{(j)}, E_T^{(j)})\}_{j=1}^m$ ($m = |\mathcal{S}| = 4$ source encoders). Writing $t_p^\star = \textsc{SellerAnchor}(x_p)$ for the seller-side anchor text and $\tilde{x} = \text{diffJPEG}_Q(x_p{+}\delta)$, the anchor-alignment loss is averaged over the source set,

$$L_{\text{ens}}(\delta) = -\frac{1}{m}\sum_{j=1}^{m} \cos\!\big(E_I^{(j)}(\tilde{x}),\, E_T^{(j)}(t_p^\star)\big), \qquad (7)$$

and the masked MI-FGSM update of (5)–(6) is applied to $\nabla_\delta L_{\text{ens}}$ without further change. Because each $E_T^{(j)}$ embeds the same seller-side anchor in its own space, δ raises anchor alignment *simultaneously* across architectures and pretraining corpora rather than overfitting the geometry of a single encoder. Ensemble optimization is a known route to adversarial transferability [13], [16], [23]; AE-CoGEO is, to our knowledge, the first to combine it with the query-free, anchor-decoupled constraint visual GEO imposes.

Source and held-out encoders are strictly disjoint: the source set is $\mathcal{S}$ = {OpenAI ViT-L/14, OpenAI ViT-B/32, EVA-02-L/14, DataComp-XL ViT-L/14} and the held-out set is {OpenCLIP-L/14, -H/14, -B/32 (all LAION-2B), SigLIP ViT-B/16}, with no shared checkpoint and no shared pretraining corpus. Every other setting — $\varepsilon = 16/255$, $T = 200$, Sobel mask, differentiable-JPEG and input-diversity environment simulation, and product-title anchor — is identical to the single-encoder CoGEO arm, so any change in held-out behavior is attributable to the consensus mechanism alone.

CoGEO and AE-CoGEO thus instrument two orthogonal axes of the same threat — the white-box and transfer ceilings — with AE-CoGEO the calibrated transfer probe; §VI-G reports the measurement.

E. Per-cohort decomposition and significance

For each adversarial image $x_p^* = x_p + \delta$ we compute four metrics against the held-out query q: *rank-lift* $\text{rank}(x_p, q) - \text{rank}(x_p^*, q)$, the positions x_p^* rises for q; *top-3 replacement rate*, the fraction of pairs where x_p^* enters the top 3 while x_p did not; *held-out* $\Delta_{\text{clip\_sim}} = \cos(E_I(x_p^*), E_T(q)) - \cos(E_I(x_p), E_T(q))$, the direct CLIP-similarity gain under q; and *SSIM* between x_p and x_p^* at the retriever's native resolution, the stealth at the regime industrial retrieval consumes. Every metric is then *decomposed* per cohort over the four ESCI relevance classes. Aggregate means across all pairs are reported alongside per-cohort means, never as a substitute. We further compute a paired Wilcoxon signed-rank test for each of the $\binom{4}{2} = 6$ method pairs on the same (q, x_p) index, so each pairwise comparison is verifiable independently of any cohort selection.

IV. Experimental Setup

We instantiate the protocol of Section III on the Amazon ESCI shopper-query benchmark and run all four attack families against the same set of product images.

A. Benchmark and sampling

We sample (q, x_p, ℓ) triples from the Amazon ESCI dataset [7] (US locale), with $\ell \in \{E, S, C, I\}$ a four-level human-graded relevance label authored by real shoppers and independent of any retriever embedding. A label-balanced sample (random seed 42) yields 491 unique product images and 560 paired triples — 125 Exact, 125 Substitute, 185 Complement, 125 Irrelevant — each resized to 224×224 [1] (CDN resolution in the released artifact). We additionally replicate the protocol on Food-101 [25] as a ranking-internal-consistency check (Fig. 2f); because its cohorts are CLIP-text-similarity quartiles over the 101 class names, they are retriever-induced and not promoted to ESCI-equivalent human-graded relevance evidence.

B. Retriever and budgets

The retriever is OpenAI CLIP ViT-L/14 [1] used identically as both the optimization target and the evaluation embedder, fixed a priori as the canonical CoGEO retriever; it also clears the pre-flight gate of (2) ($\mathrm{gap}_{E-I} = 0.0533 > 0.05$), whereas a weaker ViT-B/32 (0.0405) would not (full backbone sensitivity table in the released artifact). Cross-backbone replication uses five additional CLIP-style retrievers from the OpenCLIP toolbox [10] that vary pretraining corpus and architecture: OpenCLIP-L/14 (LAION-2B), EVA-02-L/14 [11], SigLIP ViT-B/16 [12], OpenCLIP ViT-B/32, and OpenCLIP ViT-H/14. Anchor-source robustness is measured separately in the released artifact using five VLM captioner anchors — BLIP [17], BLIP-2 [30], LLaVA-1.5 [31], Qwen2-VL [18], Qwen2.5-VL [32] — over the same six backbones (30 CoGEO cells).

For each held-out query q_i, rank-lift is computed against a fixed per-query candidate pool of size C=491 (the union of all manifest ASIN images), making it a within-pool diagnostic over the 491-image manifest rather than an absolute marketplace-scale rank: *rank* is the position of $x^\star_{p,i}$ in the descending $\cos(E_I(\cdot), E_T(q_i))$ sort. All four families operate at matched $\varepsilon = 16/255$ in L_∞, with $\varepsilon \in \{4, 8\}/255$ sweeps in Fig. 2b. CoGEO and PGD-bare run $T = 200$ momentum signed-step PGD iterations [8], [16]; CoGEO adds a differentiable JPEG layer [15] and input-diversity transforms [23], [24]. AdvCLIP trains its universal patch [4], [19], [29] over the same manifest used at evaluation, a regime favorable to AdvCLIP, so CoGEO's dominance is conservative. Co-Attack runs 200 alternating iterations at image $\varepsilon = 16/255$, text $\varepsilon_v = 0.1$ [5], its text branch perturbing a fixed seller-side anchor (*not* the held-out query). Anchor sources are held constant within method (product-title for CoGEO/PGD-bare, image-only for AdvCLIP; full hyper-parameters in the released artifact).

C. Significance protocol

For each comparison we report the paired Wilcoxon signed-rank test [9] on rank-lift, paired bootstrap 95% CIs (1000 resamples), and the held-out $\Delta_{\text{clip\_sim}}$ mean; SSIM [6] is reported at the native 224×224 input. Directional tests are one-sided; the six pairwise p-values are reported uncorrected, but the headline comparisons survive Holm–Bonferroni at the six-test count (the strongest, $p = 3.5 \times 10^{-30}$, survives any correction). The full 4×4 matrix is in the released artifact (§VI-B).

V. Main Results

Table II reports the headline four-way comparison on the full 560-pair intersection. These results measure the *white-box axis*, where optimization and scoring encoders coincide; the deployment-realistic *transfer axis* (AE-CoGEO, §III-D) is measured separately in §VI-G. Three findings emerge.

TABLE II
Four-way comparison on the 560-pair joint sample at $\varepsilon = 16/255$ with CLIP ViT-L/14. Rank-lift is the mean positions an image rises against its held-out ESCI query; SSIM is at the 224×224 input; p-values are one-sided paired Wilcoxon versus CoGEO; I→top-3 gain is against the unperturbed baseline.

Method	Rank-lift	p vs CoGEO	SSIM	I→top-3 gain
CoGEO	**19.86**	—	0.66	**+4.1 pp**
PGD-bare	12.37	1.1×10^{-4}	0.76	+2.9 pp
AdvCLIP	6.45	3.5×10^{-30}	**0.78**	+1.6 pp
Co-Attack	5.77	1.3×10^{-8}	0.76	+2.9 pp

Aggregate-mean dominance is significant: CoGEO's $+19.86$ mean rank-lift is $1.6\times$ the next-best baseline, with every pairwise paired Wilcoxon p-value well below 0.05 (to 3.5×10^{-30} versus AdvCLIP); the four-way intersection sharpens significance by including pairs on which AdvCLIP and Co-Attack both fail to move the ranking. AdvCLIP's universal patch [4] moves the ranking least — a single δ against a fixed text concept does not adapt to per-image content [19] — and Co-Attack is statistically below PGD-bare ($p = 2.4 \times 10^{-6}$); its text-anchor branch does not compensate for the loss of CoGEO's mask and momentum [16], [23], [24].

Visual stealth and retrieval lift trade off: The four SSIM values [6] rank in the opposite order from rank-lift (Table II): at 800×800 CoGEO's SSIM is ≥ 0.98, but at the 224×224 regime the retriever consumes all four fall to low SSIM (CoGEO 0.66, stealthiest AdvCLIP 0.78; Section VI-C). The right-most column of Table II reports the metric a defender ultimately cares about — pushing Irrelevant products into the top-3 a shopper sees, against a 9.1% unperturbed baseline — where CoGEO translates its largest aggregate lift into top-3 placement.

VI. Analysis

The aggregate result of §V hides a structurally heterogeneous picture, decomposed below along cohort, pairwise significance, stealth, perturbation-budget sensitivity, the Exact-cohort failure mode, cross-backbone replication, transfer, and anchor robustness, then consolidated under a bounded-under-stress synthesis (further replications in the released artifact).

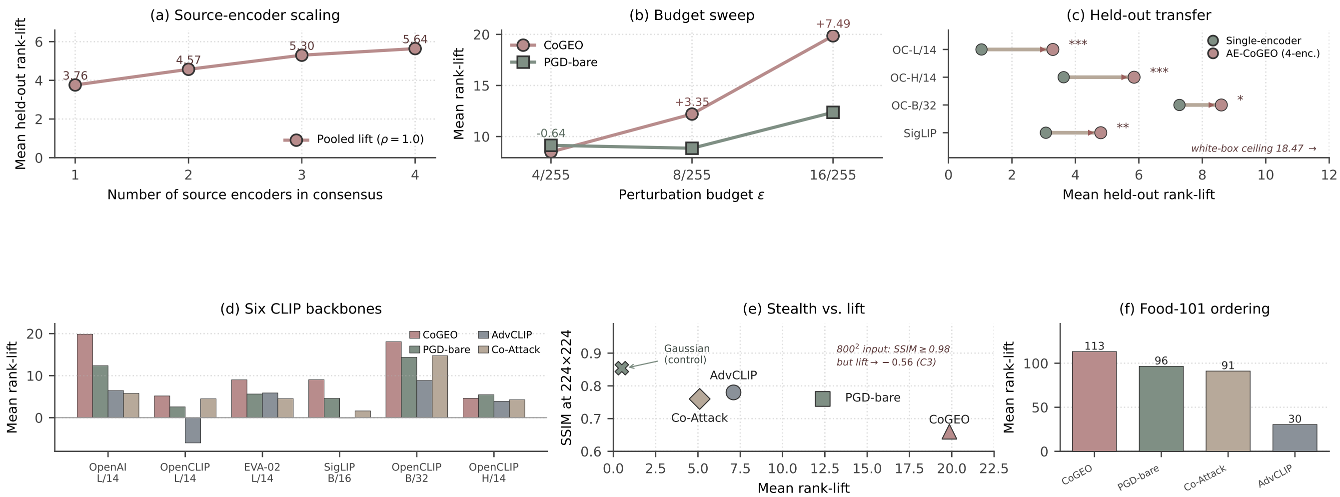

Fig. 2. AE-CoGEO transfer behavior and protocol robustness; every value matches a manuscript table. **(a)** Rank-lift scales with the source-encoder count ($3.76 \to 5.64$, $\rho = 1.0$). **(b)** Budget sweep (OpenAI ViT-L/14): the CoGEO−PGD-bare margin inverts at $\varepsilon = 4/255$ (-0.64) and grows to $+7.49$ at $16/255$. **(c)** Held-out transfer (single-encoder → four-encoder consensus, $n = 498$, Table IV): the consensus lifts all four unseen encoders (one-sided paired Wilcoxon; $^*p < 5\times10^{-2}$, $^{**}p < 10^{-2}$, $^{***}p < 10^{-3}$) but stays below the white-box ceiling (dashed). **(d)** Four-way rank-lift across six CLIP backbones ($\varepsilon = 16/255$): CoGEO is the unique top-1 on five of six. **(e)** Stealth/lift trade-off (Table II): higher rank-lift costs SSIM, all below high fidelity at 224×224 (CoGEO 0.66, AdvCLIP 0.78). **(f)** Food-101 cross-dataset check (505 triples, OpenAI ViT-L/14): the ESCI ordering CoGEO > PGD-bare > Co-Attack > AdvCLIP is preserved (gap $1.61\times \to 1.17\times$).

TABLE III

PER-COHORT MEAN RANK-LIFT OVER THE FOUR ESCI RELEVANCE CLASSES (560-PAIR, OPENAI VIT-L/14, $\varepsilon = 16/255$): THE PER-COHORT HETEROGENEITY CLAIM **(C2)**. BOLD MARKS THE PER-COHORT BEST; SIZES $n = 125$ FOR E/S/I AND 185 FOR C; VALUES MATCH TABLE II.

Method	Exact (n=125)	Substitute (n=125)	Complement (n=185)	Irrelevant (n=125)
CoGEO	3.90	5.24	**36.77**	**25.43**
PGD-bare	**7.12**	1.84	18.86	18.53
AdvCLIP	1.00	0.94	14.46	5.53
Co-Attack	6.18	**5.47**	−1.86	16.94

A. *Per-cohort rank-lift decomposition*

Table III reports the same rank-lift quantity as Table II, split across the four ESCI cohorts (sizes $n = 125$ for E/S/I and 185 for C); the split materially reshapes the headline.

Three patterns emerge. First, CoGEO's dominance is not uniform: on Exact, PGD-bare's mean exceeds CoGEO's by ~ 3.2 positions, yet matched-pair tests find no systematic per-pair advantage, so PGD-bare only *approximately matches* CoGEO there (§VI-E). Second, Complement is where CoGEO's machinery yields its largest gain ($+36.8$, $1.95\times$ PGD-bare; a small-sample peak that resolves to strict $E < S < C < I$ at $3\times$ scale, §VI-I). Third, Co-Attack alone turns negative on Complement (-1.86), its alternating image–text branch degrading ranking when the image–text correlation is already weak. The aggregate ranking thus conceals a method-specific regime: *visual GEO is a cohort-specific operator* whose lift concentrates on the harder, lower-relevance cohorts (peaking on Complement, strong on Irrelevant) while gaining no per-pair edge over PGD-bare on Exact.

B. *Pairwise paired Wilcoxon matrix*

The full 4×4 paired Wilcoxon p-value matrix (released artifact) has all but one directional comparison significant at $p < 10^{-3}$. AdvCLIP-versus-Co-Attack is the only case where the summaries disagree: AdvCLIP's higher mean (6.45 vs. 5.77) reverses under the paired test (p=4.8×10^{-7} for Co-Attack), so we adopt the rank-based reading. PGD-bare is significantly above both AdvCLIP and Co-Attack ($p < 10^{-5}$): even without CoGEO's engineering, a bare PGD recipe with an eval-independent anchor beats both better-engineered literature alternatives.

C. *SSIM versus rank-lift Pareto*

The four SSIM means in Table II rank broadly inverse to rank-lift: the methods occupy distinct points on a roughly monotonic stealth–lift curve at $\varepsilon = 16/255$, all at low SSIM at the native 224×224 input (CoGEO 0.66, stealthiest AdvCLIP 0.78; Fig. 2e). The prior CoGEO claim of SSIM ≥ 0.98 holds only at 800×800, not the regime the retriever consumes, so "high stealth and high effect" dissolves under multi-method evaluation — a trade-off structural to the budget–effect frontier, not a CoGEO-specific weakness.

D. *Perturbation-budget sensitivity*

Our main comparison fixes $\varepsilon = 16/255$, but a defender cares most about smaller budgets. Sweeping $\varepsilon \in \{4, 8, 16\}/255$ (OpenAI ViT-L/14) adds a regime-dependent qualification: at $\varepsilon = 4/255$ the headline inverts and CoGEO trails PGD-bare by 0.64, because CoGEO's auxiliary regularizers [15], [23], [24] consume a fixed share of too small a budget; as ε doubles the margin swings positive and increases across the three budgets ($+3.35$ at $8/255$, $+7.49$

at $16/255$; Fig. 2b). The stealth–lift trade-off therefore shifts *within* CoGEO as the budget changes, so a defender cannot extrapolate the $\varepsilon = 16/255$ ordering to a smaller-budget adversary.

E. Failure mode on the Exact cohort

The Exact-cohort column of Table III is striking: PGD-bare +7.12, Co-Attack +6.18, CoGEO +3.90, AdvCLIP +1.00 — two attacks that strip CoGEO's mask and environment simulation finish above it. The matched-pair tests qualify this aggregate ordering.

Of the n=125 Exact pairs, 90 tie at rank-lift 0; among the 35 non-tie pairs 17 favor PGD-bare and 18 CoGEO (sign-test p=1.00, paired Wilcoxon p=0.47; median paired difference 0, 10%-trimmed mean 0.01 — two orders below the raw 3.22 gap). The aggregate gap is thus not a systematic per-pair effect but a mechanism mismatch: the Sobel-edge mask concentrates budget on high-frequency boundaries that whole-image Exact-match pairs do not localize. We therefore do not advance "PGD-bare beats CoGEO on Exact" as a claim: the difference is within-pair noise, while other cohorts remain decisively CoGEO-favoring or balanced.

F. Cross-backbone replication

To check that the per-cohort dominance is a property of the attack family rather than of one embedding, we replicate the four-way comparison on five additional CLIP-style retrievers (§IV-B) spanning two pretraining corpora and three model scales, holding the manifest, queries, budget, iterations, and anchor fixed (Fig. 2d). This sharpens the white-box ceiling of **(C1)** into its cross-backbone form: *CoGEO is the unique top-*1 *in mean rank-lift on five of six CLIP backbones at* $\varepsilon = 16/255$*, overtaken by PGD-bare only on the highest-capacity OpenCLIP-H/14*, consistent with the Exact-cohort failure mode of §VI-E. The backbone axis is thus also a defensive surface: within the corpus-controlled LAION-2B family the manipulable band contracts monotonically with capacity ($18.07 \to 5.18 \to 4.61$, n=500, Spearman $\rho = -1.0$, Fig. 3); the n=560 OpenAI ViT-L/14 headline reaches 19.86 separately, and the all-six correlation is a non-significant $\rho = -0.64$.

External validity beyond ESCI/OpenAI: The same four-method ordering is preserved on Food-101 (Fig. 2f), so the ceiling and ordering above are properties of the attack family rather than artifacts of the ESCI/OpenAI pairing.

G. The transfer axis: significant but bounded held-out rank-lift

AE-CoGEO is the first query-free, anchor-decoupled cross-encoder probe whose role is to *measure* this transfer axis as a calibrated instrument, not to mount a leaderboard or SOTA attack. One perturbation optimized against the four-encoder source consensus is scored, without re-attack, on four held-out encoders absent from the source set, against a single-encoder CoGEO reference arm (optimized against OpenAI ViT-L/14 alone) transferred to the same encoders. The only difference

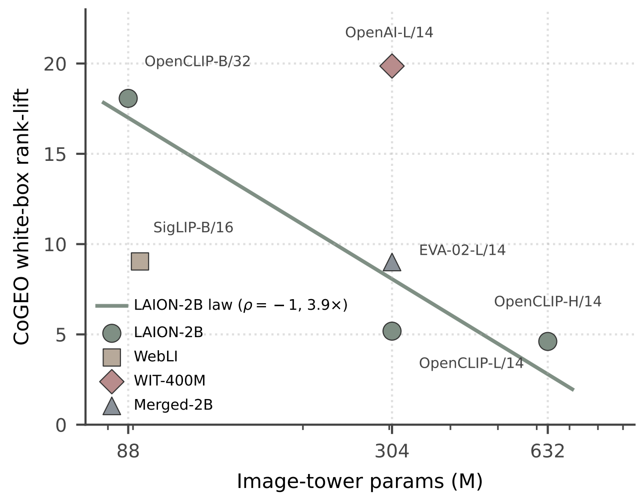

Fig. 3. Retriever capacity compresses the manipulable band. CoGEO white-box mean rank-lift versus image-tower parameter count (log scale, $\varepsilon = 16/255$, 224×224): within the corpus-controlled LAION-2B family the band contracts monotonically with capacity ($18.07 \to 5.18 \to 4.61$, Spearman $\rho = -1.0$, $\approx 3.9\times$); a larger or cleaner pretraining corpus compresses it further at fixed capacity (-74% from OpenAI WIT-400M to LAION-2B at ViT-L/14). The all-six correlation is a non-significant $\rho = -0.64$.

between arms is the source set ($n = 498$ paired ESCI pairs, $\varepsilon = 16/255$), isolating the transfer mechanism (§III-D).

TABLE IV
HELD-OUT-ENCODER TRANSFER RANK-LIFT (MEAN POSITIONS GAINED, $n = 498$, $\varepsilon = 16/255$). THE SINGLE-ENCODER ARM TRANSFERS A COGEO PERTURBATION OPTIMIZED AGAINST OPENAI VIT-L/14 ALONE; THE AE-COGEO ARM TRANSFERS ONE OPTIMIZED AGAINST THE FOUR-ENCODER SOURCE CONSENSUS (§III-D); NO HELD-OUT ENCODER APPEARS IN THE SOURCE SET. p: ONE-SIDED PAIRED WILCOXON OF AE-COGEO OVER THE SINGLE-ENCODER BASELINE. †LAION-2B-PRETRAINED.

Held-out encoder	Single-enc	AE-CoGEO	p
OpenCLIP-L/14†	1.04	**3.29**	7×10^{-5}
OpenCLIP-H/14†	3.63	**5.85**	$< 10^{-3}$
OpenCLIP-B/32†	7.28	**8.60**	0.016
SigLIP ViT-B/16	3.07	**4.80**	1.3×10^{-3}
Strongest-source white-box: **18.47** (transfer ceiling)			

Consensus transfer is real and statistically significant: The consensus lifts mean held-out rank-lift over the single-encoder baseline on all four unseen encoders, each gain significant at the per-comparison level (one-sided paired Wilcoxon, $p \in [7 \times 10^{-5}, 0.016]$; Table IV, Fig. 2c). A residual threat thus exists: an attacker ensembling public CLIP encoders measurably moves a retriever it never optimized against.

The transfer threat stays far below the white-box ceiling, across paradigms: The same source-consensus perturbation evaluated white-box on its strongest source encoder reaches the transfer ceiling of 18.47 (Table IV), two-to-six times the transferred 3.3–8.6. Scored without re-attack on a fused-encoder BLIP image–text reranker [17] that shares no encoder or objective with the source set ($n = 500$), it reaches only 7.52, still inside the held-out band (Fig. 5a): the deflation of §V survives the strongest transfer attack *and* a change

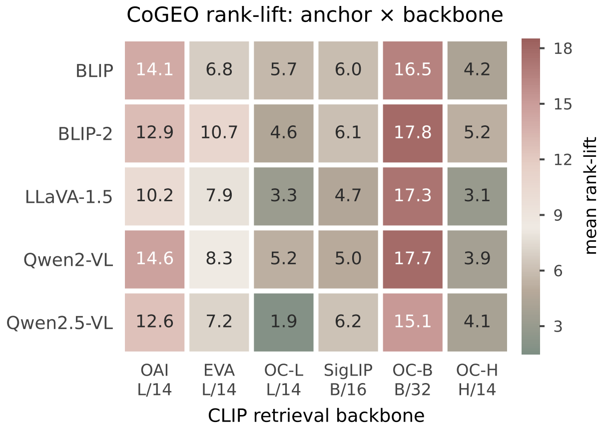

Fig. 4. CoGEO mean rank-lift across 5 VLM caption anchors (rows) × 6 CLIP backbones (columns) on the 491-ASIN ESCI manifest ($\varepsilon = 16/255$): OAI L/14, EVA (EVA-02 L/14), OC-L (OpenCLIP L/14), SigLIP (B/16), OC-B (OpenCLIP B/32), OC-H (OpenCLIP H/14). Column means span 4.1–16.9 but row means only 7.7–9.6: the retriever, not the anchor captioner, explains the bulk of the spread—the signature of an anchor-decoupled protocol.

of retrieval paradigm, and the training-free detector flags it (AUROC 0.990).

Consensus transfer scales monotonically with the source-encoder count: Growing the source set from one to four public CLIP encoders raises the pooled held-out transfer monotonically ($3.76 \rightarrow 5.64$, Spearman $\rho = 1.0$; Fig. 2a), so AE-CoGEO is a genuine ensemble-transfer instrument with a measurable scaling law.

H. Anchor-source robustness

Because the protocol requires only that the anchor be a function of x_p, we re-run CoGEO against the same six retrievers with five VLM captioners, giving the 5×6 rank-lift matrix of Fig. 4. Two facts hold: the retriever dominates the anchor (per-backbone column means span 4.1–16.9, versus only 7.7–9.6 across the five anchor captioners), and no captioner is uniformly best (within-column gap under 5 positions). Both confirm (1): any function of x_p alone is an admissible anchor, so the protocol surfaces an encoder-side property rather than a captioner artifact.

I. Bounded under stress: transfer, defense, and scale-up

The remaining robustness, transfer, and defense experiments are consolidated into one display, Figure 5 (full per-pair data and per-experiment construction in the released artifact). Three messages emerge.

The white-box rank-lift is a genuine ceiling: Every out-of-source transfer probe — a four-encoder AE-CoGEO consensus, a BLIP fused-encoder reranker, a BLIP-2 Q-Former, and a late-interaction MaxSim matcher — stays far below it (Fig. 5a), so the deflation survives a change of encoder, of retrieval paradigm, and of matching mechanism. A fifth, diffusion-prior attack family lands mid-pack (13.18, below CoGEO's 15.60 on the matched n=500 five-family panel; Table V), so CoGEO holds the top point estimate across five families — though their 95% CIs overlap and ranks 2–4 are sample-dependent (Co-Attack 12.82 vs 5.77 at 560 pairs); the per-pair distributions are strongly right-skewed for all five (Fig. 6).

TABLE V
TWO FURTHER CHECKS (OPENAI VIT-L/14, n=500). *Top*: PER-PAIR RANK-LIFT FOR FIVE WHITE-BOX FAMILIES AT ε=16/255; ALL ARE RIGHT-SKEWED (MEDIAN ≈ 1) WITH COGEO THE TOP POINT ESTIMATE (OVERLAPPING CIS). *Bottom*: TRAINING-FREE LAPLACIAN-DETECTOR AUROC BY BUDGET, ≥ 0.99 AT ε=16/255 AND DEGRADING ONLY AS THE BUDGET SHRINKS.

Five attack families (white-box), per-pair rank-lift					
Family	Mean	95% CI	Med.	% ≤ 0	Max
CoGEO	**15.60**	[10.9, 20.3]	1.0	47.0	354
PGD-bare	13.58	[8.3, 18.8]	1.0	48.2	390
Diffusion-prior	13.18	[8.5, 17.9]	1.0	48.8	388
Co-Attack	12.82	[8.2, 17.5]	1.0	48.0	348
AdvCLIP	5.06	[1.8, 8.4]	0.0	63.4	281
Training-free detector AUROC by budget					
ε=16/255	0.99–1.00		ε=8/255: 0.957–0.970		
ε=4/255	0.852–0.889		Food-101: 1.00		

The bounded operator is defender-containable, and heterogeneity is not an exploitable handle: A training-free Laplacian detector flags all four families and the AE-CoGEO transfer perturbation at AUROC ≥ 0.99, generalizing across families and to a held-out domain, degrading only as the budget shrinks (Table V); small-budget input purification removes most of the residual lift (further reduced under cohort-aware oracle allocation), and even a matched adaptive attacker stays far below the ceiling (Fig. 5b). A cohort-adaptive-budget (CAB) attacker redistributing a matched mean budget toward cheaper-to-move cohorts *under-performs* a uniform budget (13.0 vs 15.6), so heterogeneity is a diagnostic, not a handle; no tested budget is both potent and detector-evasive.

The findings are not small-sample artifacts: Rebuilding a label-balanced ESCI sample three times larger (1,430 pairs) sharpens the per-cohort gradient and the mean-shift, not per-pair directional dominance (Fig. 5c). Across all four families at this scale the mean rank-lift ordering is CoGEO 41.26 > Co-Attack 34.05 $\approx$ PGD-bare 33.38 > AdvCLIP 26.16: CoGEO has the highest mean, with Co-Attack and PGD-bare near-tied at ranks 2–3. The CoGEO$-$PGD-bare lead is $+7.89$ (bootstrap 95% CI $[2.0, 13.6]$, excluding zero) yet the paired directional Wilcoxon stays non-significant (p=0.78), so the mean advantage is tail-driven; both anchor attacks nonetheless rise strictly ($E < S < C < I$), resolving rather than reproducing the small-sample 560-pair Complement-peak ordering ($E < S < I < C$, Table III), the Exact reversal vanishing. Two directional controls reinforce this: Gaussian noise at the same budget is stealthier than every attack (SSIM 0.854) yet yields ≈ 0 rank-lift, and the same attack scored at the larger 800^2 input is more stealthy (SSIM ≥ 0.98) but loses essentially all lift, making resolution a third trade-off axis (C3; both shown in Fig. 2e).

VII. LIMITATIONS

The empirical scope is intentionally narrow. **Architecture.** Our six retrievers [1], [10]–[12] span two pretraining corpora and three scales but share the CLIP dual-encoder

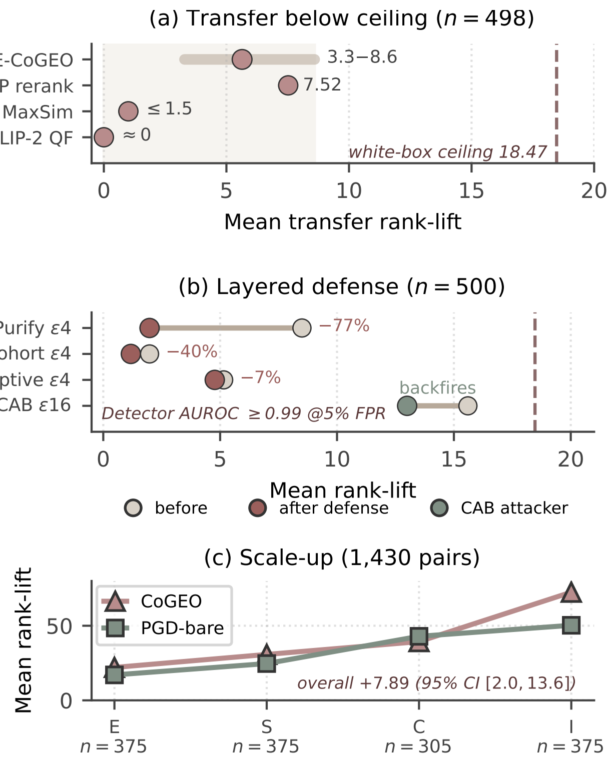

Fig. 5. Consolidated robustness, transfer, and defense evidence (rank-lift vs. the 18.47 white-box ceiling; panel (c) is an independent 1,430-pair scale-up with its own axis). **(a)** Out-of-source transfer: the AE-CoGEO four-encoder consensus (3.3–8.6), a BLIP fused-encoder reranker (7.52), a late-interaction MaxSim probe ($\lesssim 1.5$), and a BLIP-2 Q-Former (≈ 0) all sit far below the ceiling. **(b)** Layered defense at $\varepsilon{=}4/255$ (before $\rightarrow$ after): input purification ($8.50 \rightarrow 1.98$, -77%), cohort-aware oracle (-40% further), and a matched adaptive attacker ($5.15 \rightarrow 4.77$); at $\varepsilon{=}16/255$ a cohort-adaptive-budget attacker under-performs uniform (13.0 vs 15.6) and a training-free detector flags every family at AUROC ≥ 0.99. **(c)** The $3\times$ scale-up sharpens the per-cohort gradient into a strictly monotone $E<S<C<I$ for both load-bearing attacks (CoGEO−PGD $+7.89$, 95% CI $[2.0, 13.6]$). Values match §VI-I.

paradigm; §VI-G measures transfer to a BLIP fused-encoder reranker [17], which retains bounded positive transfer (mean rank-lift 7.52, within the 3.3–8.6 held-out-CLIP band), while a BLIP-2 Q-Former reranker [30] (released artifact) shows no positive transfer. A training-free late-interaction (MaxSim) probe shows the same bounded behavior; fully trained late-interaction and learned-hashing retrievers are left to future work. Within each backbone the encoder pair is both optimization target and evaluation embedder, so headline numbers are a white-box upper bound. **Resolution and budget.** Headline rankings use 224×224 at $\varepsilon = 16/255$; the released artifact reports two crossovers (800×800: CoGEO turns marginally negative, overtaken by PGD-bare; $\varepsilon = 4/255$: PGD-bare slightly exceeds CoGEO), so dominance requires a budget large enough to amortize CoGEO's auxiliary objectives. **Attack and benchmark scope.** The four-attack shortlist [4], [5], [8], [19] covers per-image gradient, universal-patch, and cross-modal families; evolutionary search is unmeasured. Crucially, the per-cohort human-relevance findings rest on ESCI's graded labels [7] alone: the Food-101 cohorts are retriever-induced (CLIP-text-similarity quartiles), not human relevance, so they probe robustness rather than substitute for graded evidence.

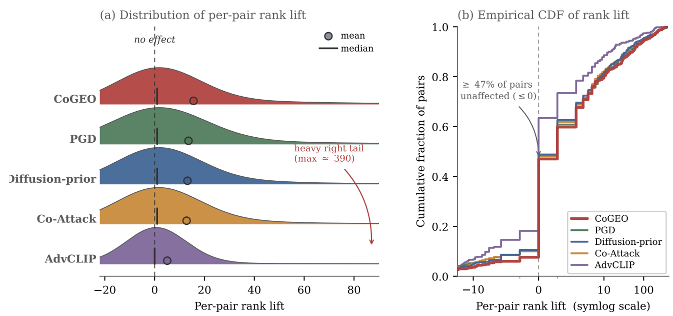

Fig. 6. Per-pair rank-lift distribution for the five white-box attack families (OpenAI ViT-L/14, $n{=}500$ each, $\varepsilon{=}16/255$). **(a)** Ridgeline densities: every family is strongly right-skewed—the typical pair barely moves (median ≈ 1; dashed "no effect" line) while a heavy right tail (up to ≈ 390) lifts the mean (open circles). **(b)** Empirical CDF on a symlog axis: the lower/further-right a curve, the heavier its tail; CoGEO is heaviest, AdvCLIP lightest, with $\geq 47\%$ of pairs unaffected ($\lesssim 0$) for every family. The headline mean thus summarises a sparse, skewed effect, not a uniform shift.

A survey of public alternatives found none simultaneously offering a second product domain, multi-level human relevance, and usable images, so this evidence stays single-domain. **Anchor independence.** The invariant of (1) closes the immediate optimization-anchor/eval-query loop but not the deeper seller-ecosystem/training-corpus loop; the VLM-anchor ablation (released artifact) bounds within-anchor variation by under 5 rank positions across five captioners and six backbones, narrowing the self-loop critique rather than erasing the per-cohort findings.

VIII. Conclusion

Findings: We recalibrate visual GEO into something a defender can act on. Within the anchor-decoupled protocol, a four-way comparison on Amazon ESCI [1], [7] across six retrievers at $\varepsilon = 16/255$ replaces optimistic self-loop scores with an honest measurement: per-image gradient optimization significantly outperforms universal-patch and cross-modal baselines (Wilcoxon $p \leq 1.1 \times 10^{-4}$), yet the per-cohort lens exposes an Exact-cohort statistical match and a monotonic stealth/lift trade-off.

Contribution 1: an honest recalibration: At its core is an evaluation methodology: an anchor$\perp$query invariant, a pre-flight retriever-discrimination gate, and a per-cohort decomposition over the four ESCI relevance classes [7]. Through the protocol, transfer, and detection results, visual GEO is not a uniform invisible threat but a cohort-specific, white-box-bounded, transfer-limited, detectable (AUROC ≥ 0.99 at $\varepsilon{=}16/255$ on a balanced benchmark) operator. Detectability and efficacy rise and fall together, so no tested budget makes visual GEO simultaneously effective and evasive (Fig. 5): a cheap input-purification knob narrows the small-budget gap, and the per-cohort heterogeneity, though not an exploitable handle, is defender-actionable through an oracle cohort allocation, with a practical cohort router left to future work.

Contribution 2: AE-CoGEO, a new cross-encoder transfer attack: To our knowledge the first of its kind for visual GEO (§III-D, VI-G), it outperforms the single-encoder transfer baseline on all four held-out encoders by a small but consistent margin ($+1.3$ to $+2.3$ positions), with a monotone source-count scaling law (Spearman $\rho = 1.0$), yet its transfer stays far below the same source-consensus white-box ceiling (3.3–8.6 versus 18.47), sharpening rather than overturning the recalibration.

Future work: Future visual-GEO claims should report stealth and rank-lift jointly at the retriever's native input size; since the 1,430-pair scale-up already sharpens the per-cohort findings into strict monotonicity, the principal remaining gap is breadth across product domains, not sample size. The protocol and both attacks ship anonymously as HONESTVGEO (https://anonymous.4open.science/r/honestvgeo-F663).

Ethics and responsible disclosure: We release AE-CoGEO and the protocol as a defensive recalibration instrument, not an operational attack, shipping the training-free detector and the input-purification knob alongside the attack. Because the evaluation runs on the public Amazon ESCI research benchmark [7] rather than any live storefront, the work targets no specific operator; we nonetheless adopt a responsible-disclosure posture and release the defense and the attack together, so that publication strengthens auditing rather than enabling undetected abuse.

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